

LankaMart Retail Performance Dashboard

An Interactive Analysis of Sales, Profitability, Fulfilment and Returns (January–June 2026)

1. Business Questions and Intended Users

Lanka Mart is an omnichannel retailer operating across Sri Lanka's nine provinces through Store, Online and Mobile App channels. Management needs a single interactive view rather than a static report, because the picture changes by region, channel, segment and time. The dashboard answers one overarching question: how is Lanka Mart performing, where are the main risks or opportunities, and what action should management consider? This splits into: (1) how are revenue and profit trending by month, category and province; (2) which channels, segments and categories are most profitable once cost is accounted for; (3) does delivery performance relate to satisfaction and returns; and (4) where is discounting eroding margin rather than growing it. The intended users are sales/merchandising managers, operations leads and finance, all of whom need to filter the same data to their own area rather than requesting separate static reports.

2. Data Quality Checks, Transformations and Calculated Fields

The supplied file LankaMart_Retail_Transactions.csv contains 721 order-line rows across 19 fields covering January to June 2026. It was loaded programmatically with pandas and inspected for structure, types, duplicates, missing values and logical consistency before any visualisation was built; no values were altered manually.

Quality issues found and how each was handled

- **One exact duplicate order** — order_id LM26-0113 appeared twice with identical values. Dropped (kept first occurrence), reducing 721 to 720 orders.
- **Inconsistent category label** — product_category contained both 'Electronics' and a lower-case 'electronic' for one row. Standardised to 'Electronics' so aggregations are not silently split across two labels.
- **Missing customer_rating** — 42 of 720 orders (5.8%) have no rating. These are genuine non-responses, not data-entry errors, so they were *not* imputed with a guessed value; they are excluded from rating averages and shown as '-' in the detail table.
- **Missing promotion** — 276 of 720 orders (38.3%) have a blank promotion field. A blank here unambiguously means no promotion was applied, so it was filled with the explicit label 'No Promotion' rather than left null, which would otherwise be dropped silently from filters.
- **Logical consistency checks** — revenue_lkr was recomputed as $\text{units} \times \text{unit_price_lkr} \times (1 - \text{discount_pct})$ and profit_lkr as $\text{revenue_lkr} - \text{cost_lkr}$ for every row; both matched exactly (zero mismatches). discount_pct (0–0.25), units (1–4) and delivery_days (0–9) are all within plausible bounds with no negative financial values.

Calculated fields

- **profit_margin_pct** = $\text{profit_lkr} \div \text{revenue_lkr} \times 100$ — makes profit comparable across orders of very different size.
- **order_month** — order_date parsed to Year-Month, used for the trend chart and date filter.
- **delivery_band** — delivery_days grouped into Fast (0–2), Standard (3–5) and Slow (6+ days), used to relate delivery speed to rating and returns without over-fitting to single-day values.

All transformations are implemented in code (`load_and_clean_data()` in `app.py`) and re-run automatically on load, so the workflow is reproducible and no chart values are hard-coded.

3. Dashboard Design Rationale

The dashboard was built with Streamlit and Plotly, using pandas for data preparation. Streamlit was chosen over Dash because its sidebar-plus-linear-flow layout matches how a manager naturally scans a report: KPIs first, then trend, then comparisons, then detail.

Chart selection

- **Time trend** — dual-line chart of monthly revenue and profit; a line encodes change over time more clearly than bars when there are only six months.
- **Categorical comparison** — horizontal bar of revenue by category, sorted so the largest is immediately visible.
- **Geographical comparison** — bar chart of revenue by province. A choropleth map was considered but rejected: with only nine provinces, a sorted bar supports faster, more accurate ranking than colour-shaded regions.
- **Relationship view** — scatter of discount level against profit margin, coloured by category, showing directly whether heavier discounting costs margin and where.
- **Distribution view** — box plot of delivery days by channel, chosen over a bar of averages because it also shows spread and outliers, which matter to an operations audience.
- **Supporting view** — bar chart of return rate by category, since returns is one of the four headline KPIs and management needs to see where it is being driven from.

Layout, colour and accessibility

KPI cards sit at the top so the four headline numbers are visible before scrolling. Charts follow a consistent twocolumn grid. One accent colour (teal) marks revenue-side charts, a second marks profit, and red is reserved specifically for the return-rate KPI and returns chart, so colour signals 'needs attention' rather than being decorative. Every chart carries hover tooltips, axis titles with units (LKR, %, days) and a title stating what it shows, so none is ambiguous alone. Category encodings pair colour with legend labels and axis position, not colour alone, keeping the dashboard readable for users with colour-vision deficiencies.

Interaction design

Four sidebar controls: an order-date range picker, and multi-select filters for province, product category and sales channel (a fifth, customer segment, is also provided). Every KPI, chart and the detail table is computed from one filtered dataframe, so a selection anywhere is reflected everywhere at once. A 'Reset all filters' button restores the full 720-order dataset in one click. The detail table lists the individual orders behind the current selection, with a live count ('Showing X of 720 total orders') confirming exactly how much data is in view.

4. Dashboard Evidence

The four views below were generated from the exact aggregation and filtering logic in `app.py` against the cleaned dataset. Figures for the full dataset and each filtered state are given in the KPI cards and captions.

Filters

Order date range

2026/01/01 – 2026/06/30

Province

Central

Eastern

North Central

North Western

Northern

Sabaragamuwa

Product category

Beauty

Electronics

Fashion

Home

Sports

Sales channel

Mobile App

Online

Store

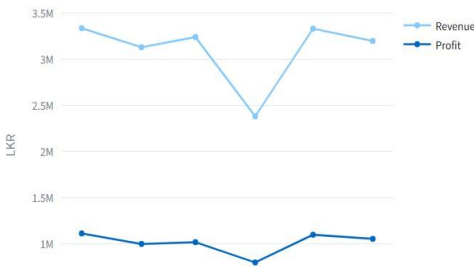
Customer segment

LankaMart Retail Performance Dashboard

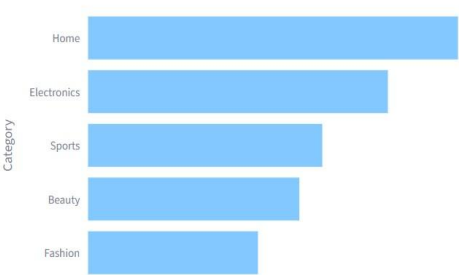
How is LankaMart performing, where are the main risks or opportunities, and what action should management consider?

Total Revenue	Total Profit	Profit Margin	Return Rate
LKR 18,621,105	LKR 6,084,741	32.7%	5.1%
			↑ 37 orders

Monthly Revenue and Profit Trend (LKR)



Revenue by Product Category (LKR)



Province

North western

Northern

Sabaragamuwa

Southern

Uva

Western

Product category

Beauty

Electronics

Fashion

Home

Sports

Sales channel

Mobile App

Online

Store

Customer segment

Consumer

Corporate

Small Business

Reset all filters

Use the controls above to explore the data.

All KPIs, charts and the detail table below respond to your selection. Click 'Reset all'

Monthly Revenue and Profit Trend (LKR)

Revenue

Profit

Revenue by Product Category (LKR)

Category

Home

Electronics

Sports

Beauty

Fashion

Revenue (LKR)

0

1M

2M

3M

4M

5M

Revenue by Province (LKR)

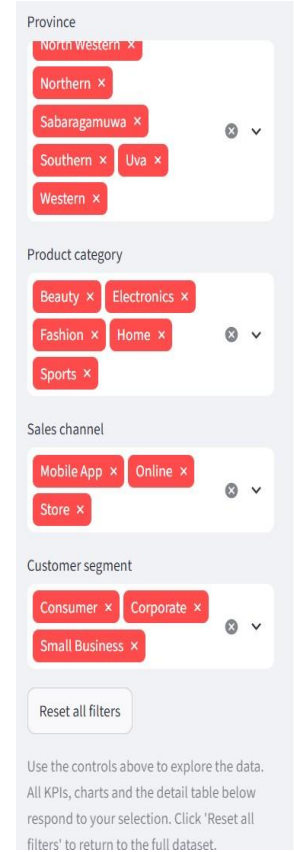
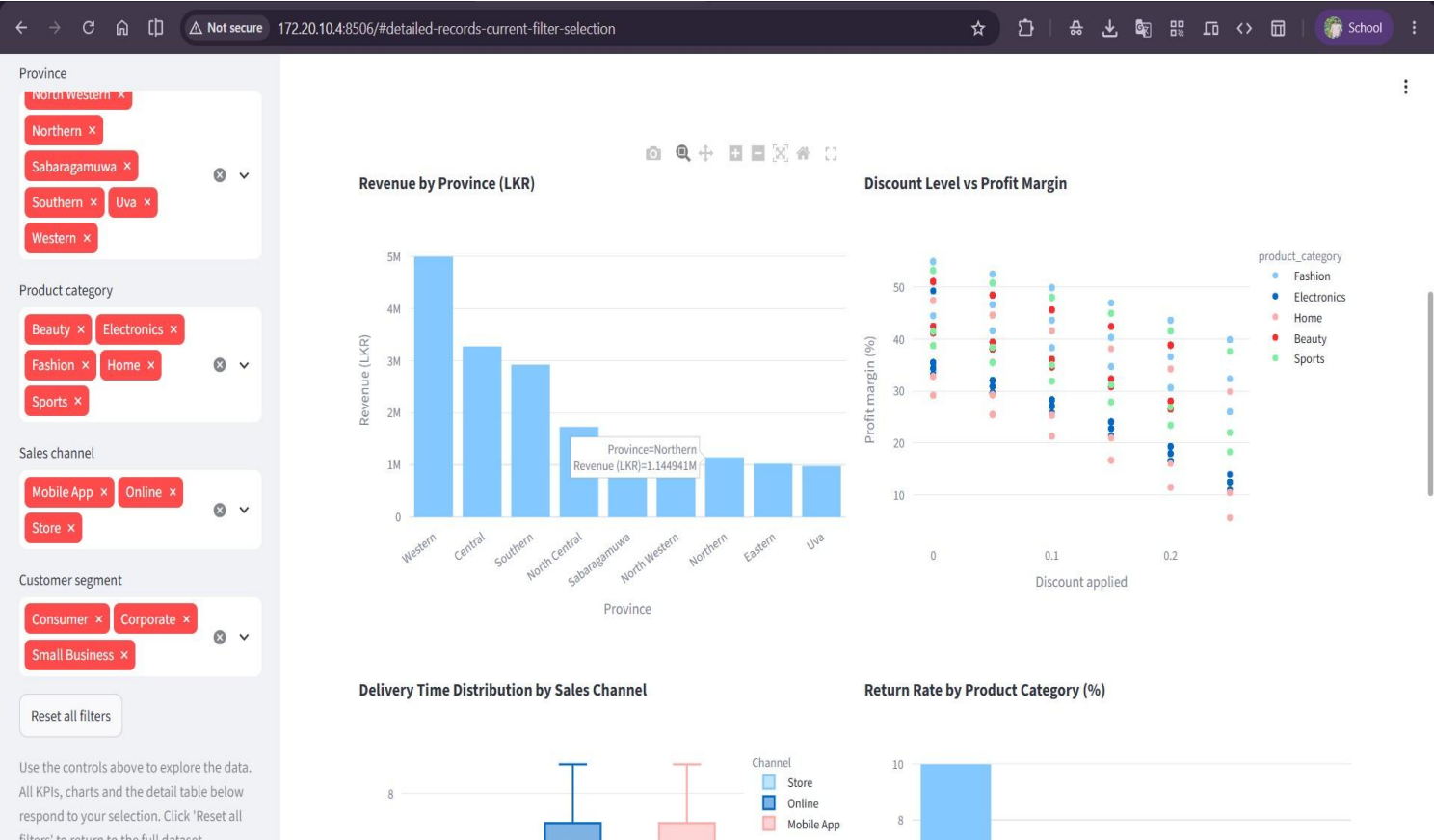
5M

Discount Level vs Profit Margin

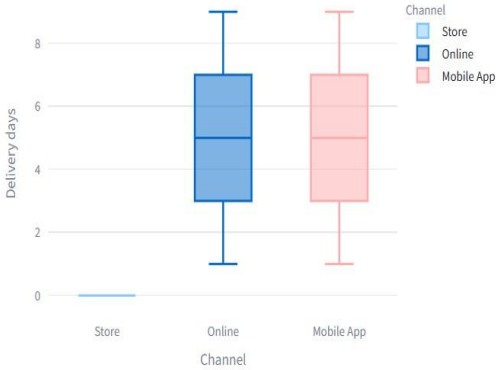
product_category

Fashion

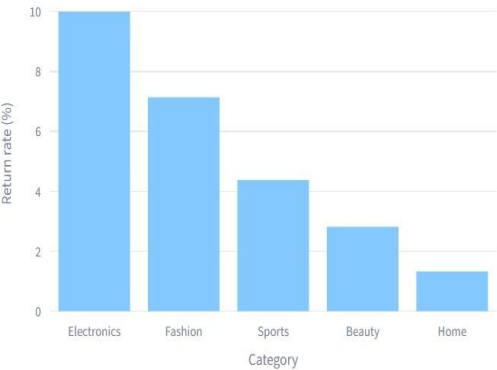
Electronics



Delivery Time Distribution by Sales Channel



Return Rate by Product Category (%)



Detailed Records (current filter selection)

	order_id	order_date	province	city	sales_channel	customer_segment	product_category	product_name	units	revenue_lkr
488	LM26-0489	2026-06-30 00:00:00	Northern	Kilinochchi	Store	Consumer	Electronics	Bluetooth Speaker	1	8991

Province

NorthernWestern ×

Northern ×

Sabarakamuwa ×

Southern ×

Uva ×

Western ×

Product category

Beauty ×

Electronics ×

Fashion ×

Home ×

Sports ×

Sales channel

Mobile App ×

Online ×

Store ×

Customer segment

Consumer ×

Corporate ×

Small Business ×

Reset all filters

Use the controls above to explore the data. All KPIs, charts and the detail table below respond to your selection. Click 'Reset all filters' to return to the full dataset.

Detailed Records (current filter selection)

	order_id	order_date	province	city	sales_channel	customer_segment	product_category	product_name	units	revenue_lkr
488	LM26-0489	2026-06-30 00:00:00	Northern	Kilinochchi	Store	Consumer	Electronics	Bluetooth Speaker	1	8991
472	LM26-0473	2026-06-30 00:00:00	Uva	Monaragala	Mobile App	Small Business	Home	Table Lamp	1	5990
181	LM26-0182	2026-06-30 00:00:00	Northern	Kilinochchi	Mobile App	Consumer	Home	Air Fryer	2	49980
115	LM26-0116	2026-06-30 00:00:00	Western	Colombo	Store	Consumer	Sports	Cricket Bat	3	38970
94	LM26-0095	2026-06-29 00:00:00	Central	Kandy	Store	Consumer	Sports	Yoga Mat	2	8980
277	LM26-0278	2026-06-29 00:00:00	Uva	Monaragala	Online	Corporate	Fashion	Casual Shirt	3	14222
361	LM26-0362	2026-06-29 00:00:00	Western	Negombo	Store	Corporate	Electronics	Wireless Earbuds	3	35073
452	LM26-0453	2026-06-28 00:00:00	North Central	Anuradhapura	Online	Consumer	Electronics	Bluetooth Speaker	2	15984

Showing 720 of 720 total orders under the current filter selection.

Key Insights (based on current filter selection)

• Home is the leading revenue category at LKR 5,360,392 in the current selection.

• Western province contributes the most revenue (LKR 4,994,615).

• Electronics has the highest return rate at 10.0%, worth investigating for quality or fit issues.

• Discount level and profit margin move together with a correlation of -0.55 — heavier discounting is associated with materially thinner margins across the

5. Findings

- **Electronics carries disproportionate return risk.** Electronics returns at 10.0% (15 of 150 orders) — double Fashion (7.1%) and nearly eight times Home (1.3%). Electronics is also LankaMart’s second-largest revenue category (LKR 4.34m), so this is a concentrated, high-value risk.
- **Delivery speed tracks directly with satisfaction.** Average customer_rating falls from 4.70 for Fast deliveries (0–2 days) to 4.05 for Standard (3–5) and 3.38 for Slow (6+ days) — over 1.3 stars from fastest to slowest. Store orders (0-day, in-person) are excluded from this comparison by definition.
- **Discounting is eroding margin, not just growing volume.** Discount level and profit margin correlate negatively at $r = -0.55$ across all 720 orders: orders at the maximum 25% discount earn substantially thinner margins, and the pattern holds within every category, not just overall.
- **Channel revenue is balanced but not equally costly to serve.** Mobile App (35.9%), Online (34.6%) and Store (29.5%) contribute similar shares of the LKR 18.6m total revenue, but Online and Mobile App average roughly 5 delivery days versus 0 for Store — so the two largest-revenue channels are the ones most exposed to the delivery-satisfaction link above.
- **Corporate orders are the least profitable segment.** Corporate generates LKR 3.05m at a 28.6% margin, versus 34.0% for Consumer and 32.1% for Small Business — plausible given typical bulk-discount terms, but worth a pricing review since Corporate is not the smallest segment.

6. Recommendations and Limitations

Recommendations

- **Investigate Electronics quality and fit before its next promotional push.** At a 10.0% return rate on a LKR 4.34m-revenue category, a targeted review of the two or three highest-return Electronics products (visible via the dashboard's category and product filters) is likely to recover more margin than further discount-driven volume growth in that category.
- **Set a maximum discount threshold by category informed by the discount-margin relationship shown in Figure 1's scatter chart**, rather than a single storewide cap, since the erosion effect is present in every category but not identical in strength across them.

Limitations

- The dataset is synthetic and covers a single six-month window (Jan–Jun 2026); seasonal effects beyond that window (for example, year-end demand) cannot be inferred from it.
- 42 orders (5.8%) have no customer rating; the reported average ratings are based only on the 94.2% of orders that were rated, and may not represent the true experience of non-responding customers if their silence is not random.

7. References

- Few, S. (2006). *Information Dashboard Design: The Effective Visual Communication of Data*. O'Reilly Media. — layout and KPI-first hierarchy principles applied in Section 3.
- Plotly Technologies Inc. Plotly Python Open Source Graphing Library. <https://plotly.com/python/> — charting library used throughout app.py.
- Streamlit Inc. Streamlit Documentation. <https://docs.streamlit.io/> — application framework used to build the interactive dashboard.